NEET (UG) – 2019 Question Paper

1. Increase in temperature of a gas filled in a container would lead to:
   (A) decrease in its pressure
   (B) decrease in intermolecular distance
   (C) increase in its mass
   (D) increase in its kinetic energy

2. Ionized hydrogen atoms and α-particles with same momenta enter perpendicular to a constant magnetic field, B. The ratio of their radii of their paths $r_H : r_\alpha$ will be:
   (A) 4 : 1 (B) 1 : 4
   (C) 2 : 1 (D) 1 : 2

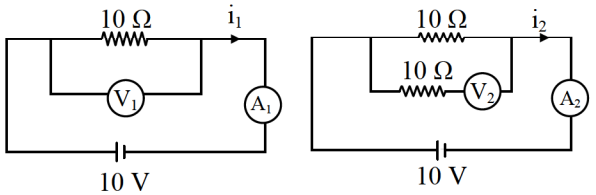
3. The speed of swimmer in still water is 20 m/s. The speed of river water is 10 m/s and is flowing due east. If he is standing on the south bank and wishes to cross the river along the shortest path, the angle at which he should make his strokes w.r.t. north is given by:
   (A) 60° west (B) 45° west
   (C) 30° west (D) 0°

4. α-particle consists of:
   (A) 2 electrons and 4 protons only
   (B) 2 protons only
   (C) 2 protons and 2 neutrons only
   (D) 2 electrons, 2 protons and 2 neutrons

5. A soap bubble, having radius of 1 mm, is blown from a detergent solution having a surface tension of 2.5×10^{-2} N/m. The pressure inside the bubble equals at a point Z_0 below the free surface of water in a container. Taking g = 10 m/s^2, density of water = 10^3 kg/m^3, the value of Z_0 is:
   (A) 1 cm (B) 0.5 cm
   (C) 100 cm (D) 10 cm

6. A mass m is attached to a thin wire and whirled in a vertical circle. The wire is most likely to break when:
   (A) the mass is at the lowest point
   (B) inclined at an angle of 60° from vertical
   (C) the mass is at the highest point
   (D) the wire is horizontal

7. In the circuits shown below, the readings of the voltmeters and the ammeters will be:



Circuit 1 **Circuit 2**

   (A) $V_1 = V_2$ and $i_1 = i_2$
   (B) $V_2 > V_1$ and $i_1 > i_2$
   (C) $V_2 > V_1$ and $i_1 = i_2$
   (D) $V_1 = V_2$ and $i_1 > i_2$

8. A force F = 20 + 10y acts on a particle in y- direction where F is in newton and y in meter. Work done by this force to move the particle from y = 0 to y = 1 m is:
   (A) 25 J (B) 20 J
   (C) 30 J (D) 5 J

9. Which of the following acts as a circuit protection device?
   (A) switch (B) fuse
   (C) conductor (D) inductor

10. The displacement of a particle executing simple harmonic motion is given by $y = A_0 + A \sin \omega t + B \cos \omega t$
    Then the amplitude of its oscillation is given by:
    (A) $\sqrt{A_0^2 + (A+B)^2}$ (B) A + B
    (C) $A_0 + \sqrt{A^2 + B^2}$ (D) $\sqrt{A^2 + B^2}$

11. At a point A on the earth's surface the angle of dip, δ = +25°. At a point B on the earth's surface the angle of dip, δ = –25°. We can interpret that:
    (A) A is located in the northern hemisphere and B is located in the southern hemisphere.
    (B) A and B are both located in the southern hemisphere.
    (C) A and B are both located in the northern hemisphere.
    (D) A is located in the southern hemisphere and B is located in the northern hemisphere.

12. Body A of mass 4m moving with speed u collides with another body B of mass 2m, at rest. The collision is head on and elastic in nature. After the collision the fraction of energy lost by the colliding body A is:
    (A) $\frac{4}{9}$ (B) $\frac{5}{9}$ (C) $\frac{1}{9}$ (D) $\frac{8}{9}$

13. Pick the wrong answer in the context with rainbow.
    (A) An observer can see a rainbow when his front is towards the sun.
    (B) Rainbow is a combined effect of dispersion, refraction and reflection of sunlight.

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(C) When the light rays undergo two internal reflections in a water drop, a secondary rainbow is formed.
(D) The order of colours is reversed in the secondary rainbow.

14. A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop after 2π revolutions is
(A) 12×10^{-4} Nm (B) 2×10^{6} Nm
(C) 2×10^{-6} Nm (D) 2×10^{-3} Nm

15. For a p-type semiconductor, which of the following statements is true?
(A) Holes are the majority carriers and pentavalent atoms are the dopants.
(B) Electrons are the majority carriers and pentavalent atoms are the dopants.
(C) Electrons are the majority carriers and trivalent atoms are the dopants.
(D) Holes are the majority carriers and trivalent atoms are the dopants.

16. An electron is accelerated through a potential difference of 10,000 V. Its de Broglie wavelength is, (nearly) ($m_e = 9 \times 10^{-31}$ kg)
(A) 12.2×10^{-14} m (B) 12.2 nm
(C) 12.2×10^{-13} m (D) 12.2×10^{-12} m

17. The work done to raise a mass m from the surface of the earth to a height h, which is equal to the radius of the earth, is:
(A) $\frac{1}{2}$ mgR (B) $\frac{3}{2}$ mgR
(C) mgR (D) 2mgR

18. In total internal reflection when the angle of incidence is equal to the critical angle for the pair of media in contact, what will be angle of refraction?
(A) equal to angle of incidence
(B) 90°
(C) 180°
(D) 0°

19. When a block of mass M is suspended by a long wire of length L, the length of the wire becomes (L + *l*). The elastic potential energy stored in the extended wire is:
(A) $\frac{1}{2}$ Mg*l* (B) $\frac{1}{2}$ MgL
(C) Mg*l* (D) MgL

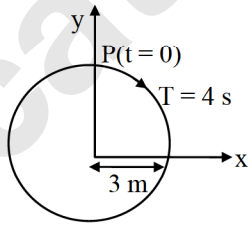
20. Which colour of the light has the longest wavelength?
(A) green (B) violet
(C) red (D) blue

21. A hollow metal sphere of radius R is uniformly charged. The electric field due to the sphere at a distance r from the centre:
(A) Zero as r increases for r < R, increases as r increases for r > R
(B) decreases as r increases for r < R and for r > R
(C) increases as r increases for r < R and for r > R
(D) zero as r increases for r < R, decreases as r increases for r > R

22. The unit of thermal conductivity is:
(A) WmK^{-1} (B) $Wm^{-1}K^{-1}$
(C) JmK^{-1} (D) $Jm^{-1}K^{-1}$

23. In which of the following processes, heat is neither absorbed nor released by a system?
(A) isobaric (B) isochoric
(C) isothermal (D) adiabatic

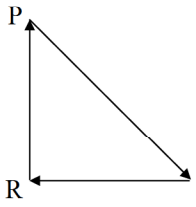
24. The radius of circle, the period of revolution, initial position and sense of revolution are indicated in the figure.



y-projection of the radius vector of rotating particle P is:
(A) $y(t) = 3\cos\left(\frac{3\pi t}{2}\right)$, where y in m
(B) $y(t) = 3\cos\left(\frac{\pi t}{2}\right)$, where y in m
(C) $y(t) = -3\cos 2\pi t$, where y in m
(D) $y(t) = 4\sin\left(\frac{\pi t}{2}\right)$, where y in m

25. Two parallel infinite line charges with linear charge densities + λ C/m and – λ C/m are placed at a distance of 2R in free space. What is the electric field mid-way between the two line charges?
(A) $\frac{\lambda}{\pi\varepsilon_0 R}$ N/C (B) $\frac{\lambda}{2\pi\varepsilon_0 R}$ N/C
(C) zero (D) $\frac{2\lambda}{\pi\varepsilon_0 R}$ N/C

26. A particle moving with velocity $\vec{v}$ is acted by three forces shown by the vector triangle PQR. The velocity of the particle will:



(A) remain constant
(B) change according to the smallest force $\overrightarrow{QR}$
(C) increase
(D) decrease

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27. Two point charges A and B, having charges + Q and – Q respectively, are placed at certain distance apart and force acting between them is F. If 25% charge of A is transferred to B, then force between the charges becomes:

(A) $\frac{16F}{9}$ (B) $\frac{4F}{3}$ (C) F (D) $\frac{9F}{16}$

28. Average velocity of a particle executing SHM in one complete vibration is:

(A) $\frac{A\omega^2}{2}$ (B) zero (C) $\frac{A\omega}{2}$ (D) $A\omega$

29. In an experiment, the percentage of error occurred in the measurement of physical quantities A, B, C and D are 1%, 2%, 3% and 4% respectively. Then the maximum percentage of error in the measurement X, where $X = \frac{A^2 B^{\frac{1}{2}}}{C^{\frac{1}{3}} D^3}$, will be:

(A) –10 % (B) 10 %

(C) $\left(\frac{3}{13}\right)$ % (D) 16 %

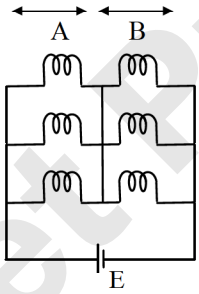
30. Six similar bulbs are connected as shown in the figure with a DC source of emf E, and zero internal resistance.
The ratio of power consumption by the bulbs when (i) all are glowing and (ii) in the situation when two from section A and one from section B are glowing, will be

(A) 1: 2

(B) 2 : 1

(C) 4 : 9

(D) 9 : 4

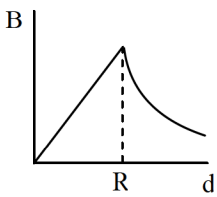


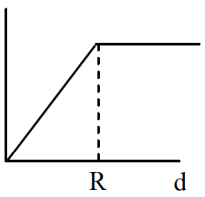
31. Two particles A and B are moving in uniform circular motion in concentric circles of radii r_A and r_B with speed v_A and v_B respectively. Their time period of rotation is the same. The ratio of angular speed of A to that of B will be:

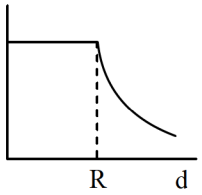
(A) $r_B : r_A$ (B) 1 : 1

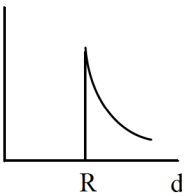
(C) $r_A : r_B$ (D) $v_A : v_B$

32. A cylindrical conductor of radius R is carrying a constant current. The plot of the magnitude of the magnetic field, B with the distance, d, from the centre of the conductor, is correctly represented by the figure:

(A) 

(B) 

(C) 

(D) 

33. When an object is shot from the bottom of a long smooth inclined plane kept at an angle 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object is shot with the same velocity, it can travel x_2 distance. Then $x_1 : x_2$ will be :

(A) $1 : \sqrt{3}$ (B) $1 : 2\sqrt{3}$

(C) $1 : \sqrt{2}$ (D) $\sqrt{2} : 1$

34. A 800 turn coil of effective area 0.05 m^2 is kept perpendicular to a magnetic field 5×10^{-5} T. When the plane of the coil is rotated by 90° around any of its coplanar axis in 0.1 s, the e.m.f. induced in the coil will be:

(A) 2×10^{-3} V (B) 0.02 V

(C) 2 V (D) 0.2 V

35. A copper rod of 88 cm and an aluminium rod of unknown length have their increase in length independent of increase in temperature. The length of aluminium rod is:
($\alpha_{Cu} = 1.7 \times 10^{-5}\ K^{-1}$ and $\alpha_{Al} = 2.2 \times 10^{-5}\ K^{-1}$)

(A) 88 cm (B) 68 cm

(C) 6.8 cm (D) 113.9 cm

36. In which of the following devices, the eddy current effect is not used?

(A) electromagnet
(B) electric heater
(C) induction furnace
(D) magnetic braking in train

37. A parallel plate capacitor of capacitance 20 μF is being charged by a voltage source whose potential is changing at the rate of 3V/s. The conduction current through the connecting wires, and the displacement current through the plates of the capacitor, would be, respectively:

(A) 60 μA, zero (B) zero, zero

(C) zero, 60 μA (D) 60 μA, 60 μA

38. Two similar thin equiconvex lenses, of focal length f each, are kept coaxially in contact with each other such that the focal length of the combination is F_1. When the space between the two lenses is filled with glycerin (which has the same refractive index (μ = 1.5) as that of glass) then the equivalent focal length is F_2. The ratio $F_1 : F_2$ will be:

(A) 2 : 3 (B) 3 : 4

(C) 2 : 1 (D) 1 : 2

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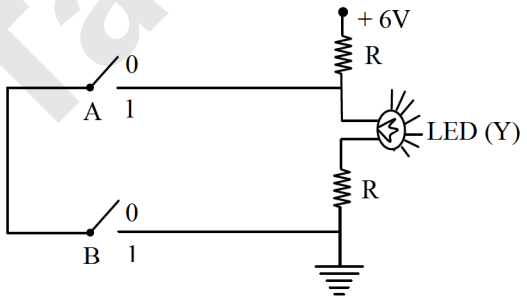
39. A block of mass 10 kg is in contact against the inner wall of a hollow cylindrical drum of radius 1 m. The coefficient of friction between the block and the inner wall of the cylinder is 0.1. The minimum angular velocity needed for the cylinder to keep the block stationary when the cylinder is vertical and rotating about its axis, will be ($g = 10 m/s^2$)
(A) 10 rad/s (B) 10π rad/s
(C) $\sqrt{10}$ rad/s (D) $\frac{10}{2\pi}$ rad/s

40. In a double slit experiment, when light of wavelength 400 nm was used, the angular width of the first minima formed on a screen placed 1 m away, was found to be 0.2°. What will be the angular width of the first minima, if the entire experimental apparatus is immersed in water? ($\mu_{water} = 4/3$)
(A) 0.05° (B) 0.1°
(C) 0.266° (D) 0.15°

41. A disc of radius 2 m and mass 100 kg rolls on a horizontal floor. Its centre of mass has speed of 20 cm/s. How much work is needed to stop it?
(A) 2 J (B) 1 J
(C) 3 J (D) 30 kJ

42. The total energy of an electron in an atom in an orbit is –3.4 eV. Its kinetic and potential energies are, respectively
(A) 3.4 eV, – 6.8 eV
(B) 3.4 eV, 3.4 eV
(C) –3.4 eV, –3.4 eV
(D) –3.4 eV, – 6.8 eV

43. A small hole of area of cross-section 2 mm^2 is present near the bottom of a fully filled open tank of height 2 m. Taking $g = 10 m/s^2$, the rate of flow of water through the open hole would be nearly:
(A) 2.23×10^{-6} m^3/s
(B) 6.4×10^{-6} m^3/s
(C) 12.6×10^{-6} m^3/s
(D) 8.9×10^{-6} m^3/s

44.


The correct Boolean operation represented by the circuit diagram drawn is:
(A) NAND (B) NOR
(C) AND (D) OR

45. A body weighs 200 N on the surface of the earth. How much will it weigh half way down to the centre of the earth?
(A) 250 N (B) 100 N
(C) 150 N (D) 200 N

46. In which case change in entropy is negative?
(A) Expansion of a gas at constant temperature
(B) Sublimation of solid to gas
(C) $2H_{(g)} \longrightarrow H_{2(g)}$
(D) Evaporation of water

47. The method used to remove temporary hardness of water is _____.
(A) Clark's method
(B) ion-exchange method
(C) synthetic resins method
(D) Calgon's method

48. For the chemical reaction,
$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$
the CORRECT option is _____.
(A) $-\frac{d[N_2]}{dt} = 2\frac{d[NH_3]}{dt}$ (B) $-\frac{d[N_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$
(C) $3\frac{d[H_2]}{dt} = 2\frac{d[NH_3]}{dt}$ (D) $-\frac{1}{3}\frac{d[H_2]}{dt} = -\frac{1}{2}\frac{d[NH_3]}{dt}$

49. The number of sigma (σ) and pi (π) bonds in pent-2-en-4-yne is
(A) 8 σ bonds and 5π bonds
(B) 11 σ bonds and 2 π bonds
(C) 13 σ bonds and no π bonds
(D) 10 σ bonds and 3 π bonds

50. Which of the following reactions are disproportionation reactions?
(a) $2Cu^+ \longrightarrow Cu^{2+} + Cu^0$
(b) $3MnO_4^{2-} + 4H^+ \longrightarrow 2MnO_4^- + MnO_2 + 2H_2O$
(c) $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$
(d) $2MnO_4^- + 3Mn^{2+} + 2H_2O \longrightarrow 5MnO_2 + 4H^+$
(A) (a), (b) and (c) (B) (a), (c) and (d)
(C) (a) and (d) only (D) (a) and (b) only

51. The CORRECT order of the basic strength of methyl substituted amines in aqueous solution is _____.
(A) $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH$
(B) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2$
(C) $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$
(D) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$

52. Identify the INCORRECT statement related to PCl_5 from the following.
(A) Two axial P – Cl bonds make an angle of 180° with each other.
(B) Axial P – Cl bonds are longer than equatorial P – Cl bonds.
(C) PCl_5 molecule is non-reactive.
(D) Three equatorial P – Cl bonds make an angle of 120° with each other.

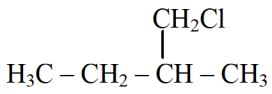
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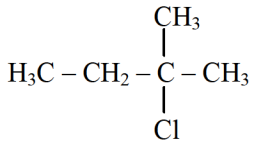
53. Which of the following series of transitions in the spectrum of hydrogen atom falls in visible region?
(A) Balmer series (B) Paschen series
(C) Brackett series (D) Lyman series

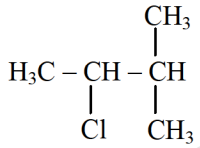
54. For a cell involving one electron $E^{\circ}_{cell} = 0.59$ V at 298 K, the equilibrium constant for the cell reaction is _____.
[Given that $\frac{2.303\,RT}{F} = 0.059$ V at T = 298 K]
(A) 1.0×10^{5} (B) 1.0×10^{10}
(C) 1.0×10^{30} (D) 1.0×10^{2}

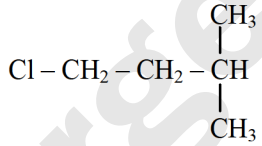
55. Among the following, the narrow spectrum antibiotic is _____.
(A) ampicillin (B) amoxycillin
(C) chloramphenicol (D) penicillin G

56. An alkene 'A' on reaction with O_3 and $Zn\text{-}H_2O$ gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene 'A' gives 'B' as the major product. The structure of product 'B' is _____.

(A) 

(B) 

(C) 

(D) 

57. Which mixture of the solutions will lead to the formation of negatively charged colloidal $[AgI]I^-$ sol?
(A) 50 mL of 1 M $AgNO_3$ + 50 mL of 2 M KI
(B) 50 mL of 2 M $AgNO_3$ + 50 mL of 1.5 M KI
(C) 50 mL of 0.1 M $AgNO_3$ + 50 mL of 0.1 M KI
(D) 50 mL of 1 M $AgNO_3$ + 50 mL of 1.5 M KI

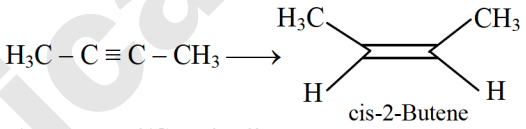
58. Conjugate bases for Bronsted acids H_2O and HF are _____.
(A) H_3O^+ and F^-, respectively
(B) OH^- and F^-, respectively
(C) H_3O^+ and H_2F^+, respectively
(D) OH^- and H_2F^+, respectively

59. Which of the following is an amphoteric hydroxide?
(A) $Ca(OH)_2$ (B) $Mg(OH)_2$
(C) $Be(OH)_2$ (D) $Sr(OH)_2$

60. What is the CORRECT electronic configuration of the central atom in $K_4[Fe(CN)_6]$ based on crystal field theory?
(A) $t_{2g}^6\, e_g^0$ (B) $e^3\, t_2^3$
(C) $e^4\, t_2^2$ (D) $t_{2g}^4\, e_g^2$

61. A compound is formed by cation C and anion A. The anions form hexagonal close packed (hcp) lattice and the cations occupy 75% of octahedral voids. The formula of the compound is _____.
(A) C_3A_2 (B) C_3A_4
(C) C_4A_3 (D) C_2A_3

62. The most suitable reagent for the following conversion is _____.



cis-2-Butene

(A) H_2, Pd/C, quinoline
(B) Zn/HCl
(C) Hg^{2+}/H^+, H_2O
(D) Na/liquid NH_3

63. The biodegradable polymer is _____.
(A) nylon-2-nylon 6 (B) nylon-6
(C) buna-S (D) nylon-6,6

64. For the second period elements the CORRECT increasing order of first ionization enthalpy is _____.
(A) Li < B < Be < C < O < N < F < Ne
(B) Li < B < Be < C < N < O < F < Ne
(C) Li < Be < B < C < O < N < F < Ne
(D) Li < Be < B < C < N < O < F < Ne

65. Which will make basic buffer?
(A) 100 mL of 0.1 M CH_3COOH + 100 mL of 0.1 M NaOH
(B) 100 mL of 0.1 M HCl + 200 mL of 0.1 M NH_4OH
(C) 100 mL of 0.1 M HCl + 100 mL of 0.1 M NaOH
(D) 50 mL of 0.1 M NaOH + 25 mL of 0.1 M CH_3COOH

66. Which is the CORRECT thermal stability order for H_2E (E = O, S, Se, Te and Po)?
(A) $H_2O < H_2S < H_2Se < H_2Te < H_2Po$
(B) $H_2Po < H_2Te < H_2Se < H_2S < H_2O$
(C) $H_2Se < H_2Te < H_2Po < H_2O < H_2S$
(D) $H_2S < H_2O < H_2Se < H_2Te < H_2Po$

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67. Under isothermal condition, a gas at 300 K expands from 0.1 L to 0.25 L against a constant external pressure of 2 bar. The work done by the gas is _____.
(Given that 1 L bar = 100 J)
(A) 5 kJ (B) 25 J
(C) 30 J (D) –30 J

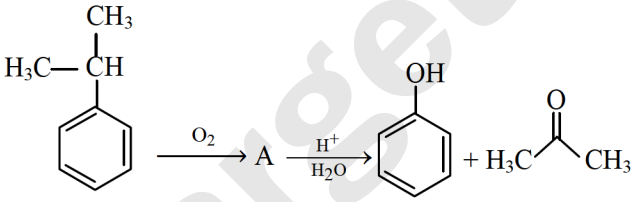
68. Among the following, the one that is NOT a greenhouse gas is _____.
(A) methane (B) ozone
(C) sulphur dioxide (D) nitrous oxide

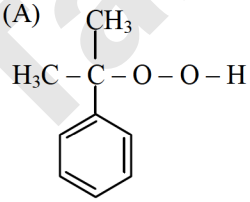
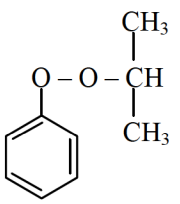
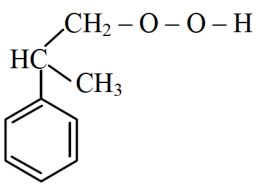
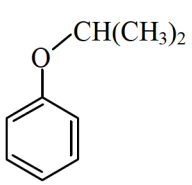
69. For an ideal solution, the CORRECT option is _____.
(A) $\Delta_{mix}V \neq 0$ at constant T and P
(B) $\Delta_{mix}H = 0$ at constant T and P
(C) $\Delta_{mix}G = 0$ at constant T and P
(D) $\Delta_{mix}S = 0$ at constant T and P

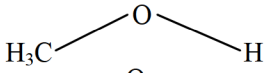
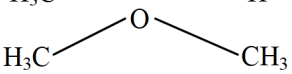
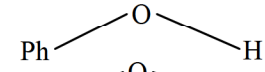
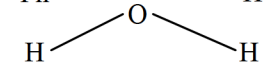
70. A gas at 350 K and 15 bar has molar volume 20 percent smaller than that for an ideal gas under the same conditions. The CORRECT option about the gas and its compressibility factor (Z) is _____.
(A) Z > 1 and repulsive forces are dominant
(B) Z < 1 and attractive forces are dominant
(C) Z < 1 and repulsive forces are dominant
(D) Z > 1 and attractive forces are dominant

71. Which one is malachite from the following?
(A) $Cu(OH)_2$ (B) Fe_3O_4
(C) $CuCO_3.Cu(OH)_2$ (D) $CuFeS_2$

72. The structure of intermediate A in the following reaction, is _____.



(A) 
(B) 
(C) 
(D) 

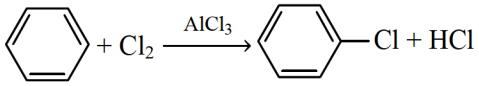
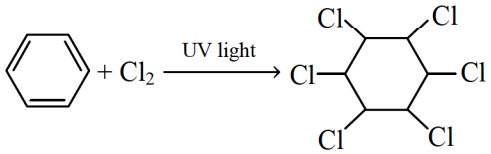
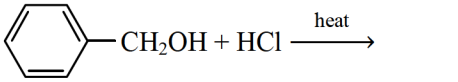
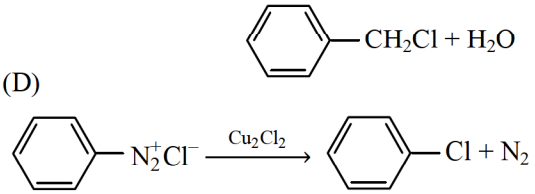
73. The compound that is most difficult to protonate is _____.
(A) 
(B) 
(C) 
(D) 

74. The manganate and permanganate ions are tetrahedral, due to _____.
(A) there is no π-bonding
(B) the π-bonding involves overlap of p-orbitals of oxygen with p-orbitals of manganese
(C) the π-bonding involves overlap of d-orbitals of oxygen with d-orbitals of manganese
(D) the π-bonding involves overlap of p-orbitals of oxygen with d-orbitals of manganese

75. Which of the following species is NOT stable?
(A) $[GeCl_6]^{2-}$ (B) $[Sn(OH)_6]^{2-}$
(C) $[SiCl_6]^{2-}$ (D) $[SiF_6]^{2-}$

76. The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is _____.
(A) 20 (B) 30 (C) 40 (D) 10

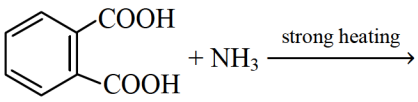
77. If the rate constant for a first order reaction is k, the time (t) required for the completion of 99% of the reaction is given by _____.
(A) t = 6.909/k (B) t = 4.606/k
(C) t = 2.303/k (D) t = 0.693/k

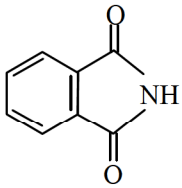
78. Among the following, the reaction that proceeds through an electrophilic substitution is _____.
(A) 
(B) 
(C) 
(D) 

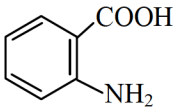
Target Publications

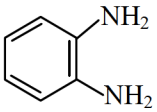
79. Enzymes that utilize ATP in phosphate transfer require an alkaline earth metal (M) as the cofactor. M is _____.
(A) Mg (B) Ca
(C) Sr (D) Be

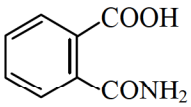
80. The major product of the following reaction is _____.



(A) 

(B) 

(C) 

(D) 

81. pH of a saturated solution of $Ca(OH)_2$ is 9. The solubility product (K_{sp}) of $Ca(OH)_2$ is _____.
(A) 0.25×10^{-10} (B) 0.125×10^{-15}
(C) 0.5×10^{-10} (D) 0.5×10^{-15}

82. Which of the following diatomic molecular species has only π bonds according to molecular orbital theory?
(A) N_2 (B) C_2
(C) Be_2 (D) O_2

83. Which of the following is INCORRECT statement?
(A) $SiCl_4$ is easily hydrolyzed.
(B) GeX_4 (X = F, Cl, Br, I) is more stable than GeX_2.
(C) SnF_4 is ionic in nature.
(D) PbF_4 is covalent in nature.

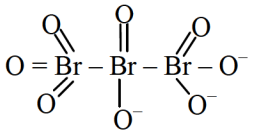
84. 4d, 5p, 5f and 6p orbitals are arranged in the order of decreasing energy. The CORRECT option is _____.
(A) 6p > 5f > 5p > 4d
(B) 6p > 5f > 4d > 5p
(C) 5f > 6p > 4d > 5p
(D) 5f > 6p > 5p > 4d

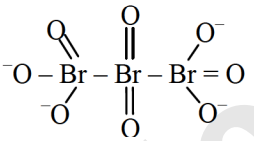
85. Match the xenon compounds in column-I with its structure in column-II and assign the CORRECT code:

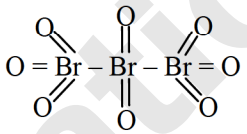
	Column-I		Column-II
i.	XeF_4	a.	Pyramidal
ii.	XeF_6	b.	Square planar
iii.	$XeOF_4$	c.	Distorted octahedral
iv.	XeO_3	d.	Square pyramidal

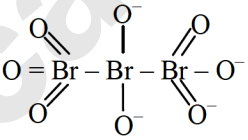
(A) i – b, ii – c, iii – d, iv – a
(B) i – b, ii – c, iii – a, iv – d
(C) i – c, ii – d, iii – a, iv – b
(D) i – a, ii – b, iii – c, iv – d

86. The CORRECT structure of tribromooctaoxide is _____.

(A) 

(B) 

(C) 

(D) 

87. The mixture that forms maximum boiling azeotrope is _____.
(A) ethanol + water
(B) acetone + carbon disulphide
(C) heptane + octane
(D) water + nitric acid

88. The non-essential amino acid among the following is _____.
(A) leucine (B) alanine
(C) lysine (D) valine

89. For the cell reaction,
$2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \longrightarrow 2Fe^{2+}_{(aq)} + I_{2(aq)}$
E°_{cell} = 0.24 V at 298 K. The standard Gibbs energy ($\Delta_r G^\circ$) of the cell reaction is _____.
[Given that Faraday constant F = 96500 C mol^{-1}]
(A) –23.16 kJ mol^{-1} (B) 46.32 kJ mol^{-1}
(C) 23.16 kJ mol^{-1} (D) –46.32 kJ mol^{-1}

90. Match the following:

	Column-I		Column-II
i.	Pure nitrogen	a.	Chlorine
ii.	Haber's process	b.	Sulphuric acid
iii.	Contact process	c.	Ammonia
iv.	Deacon's process	d.	Sodium azide or Barium azide

(A) i – b, ii – d, iii – a, iv – c
(B) i – c, ii – d, iii – b, iv – a
(C) i – d, ii – c, iii – b, iv – a
(D) i – a, ii – b, iii – c, iv – d

Target Publications

91. Select the correctly written scientific name of Mango which was first described by Carolus Linnaeus:
(A) *Mangifera indica*
(B) *Mangifera Indica*
(C) *Mangifera indica* Car. Linn.
(D) *Mangifera indica* Linn

92. In some plants, the female gamete develops into embryo without fertilization. This phenomenon is known as:
(A) syngamy (B) parthenogenesis
(C) autogamy (D) parthenocarpy

93. Which of the following statements is INCORRECT?
(A) Conidia are produced exogenously and ascospores endogenously.
(B) Yeasts have filamentous bodies with long thread - like hyphae.
(C) Morels and truffles are edible delicacies.
(D) *Claviceps* is a source of many alkaloids and LSD.

94. What is the fate of the male gametes discharged in the synergid?
(A) One fuses with the egg, other(s) fuse(s) with synergid nucleus.
(B) One fuses with the egg and other fuses with central cell nuclei.
(C) One fuses with the egg, other(s) degenerate(s) in the synergid.
(D) All fuse with the egg.

95. From evolutionary point of view, retention of the female gametophyte with developing young embryo on the parent sporophyte for some time, is first observed in:
(A) Pteridophytes (B) Gymnosperms
(C) Liverworts (D) Mosses

96. Select the CORRECT sequence for transport of sperm cells in male reproductive system.
(A) Seminiferous tubules → Vasa efferentia → Epididymis → Inguinal canal → Urethra
(B) Testis → Epididymis → Vasa efferentia → Vas deferens → Ejaculatory duct → Inguinal canal → Urethra → Urethral meatus
(C) Testis → Epididymis → Vasa efferentia › Rete testis › Inguinal canal › Urethra
(D) Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra → Urethral meatus

97. Match the following organisms with their respective characteristics:

i.	*Pila*	p.	Flame cells
ii.	*Bombyx*	q.	Comb plates
iii.	*Pleurobrachia*	r.	Radula
iv.	*Taenia*	s.	Malpighian tubules

Select the correct option from the following:
(A) i – q, ii – s, iii – r, iv – p
(B) i – r, ii – q, iii – s, iv – p
(C) i – r, ii – q, iii – p, iv – s
(D) i – r, ii – s, iii – q, iv – p

98. Select the hormone releasing Intra Uterine Devices.
(A) Progestasert, LNG - 20
(B) Lippes Loop, Multiload 375
(C) Vaults, LNG - 20
(D) Multiload 375, Progestasert

99. Placentation in which ovules develop on the inner wall of the ovary or in peripheral part is:
(A) Parietal (B) Free central
(C) Basal (D) Axile

100. What map unit (Centimorgan) is adopted in the construction of genetic maps?
(A) A unit of distance between two expressed genes on chromosomes, representing 1% cross over.
(B) A unit of distance between genes on chromosomes, representing 50% cross over.
(C) A unit of distance between two expressed genes, representing 10% cross over.
(D) A unit of distance between two expressed genes, representing 100% cross over.

101. Which one of the following equipments is essentially required for growing microbes on a large scale, for industrial production of enzymes?
(A) Industrial oven (B) Bioreactor
(C) BOD incubator (D) Sludge digester

102. Phloem in gymnosperms lacks:
(A) companion cells only
(B) both sieve tubes and companion cells
(C) albuminous cells and sieve cells
(D) sieve tubes only

103. Purines found both in DNA and RNA are:
(A) Guanine and cytosine
(B) Cytosine and thymine
(C) Adenine and thymine
(D) Adenine and guanine

104. The ciliated epithelial cells are required to move particles or mucus in a specific direction. In humans, these cells are mainly present in:
(A) Eustachian tube and salivary duct
(B) Bronchioles and fallopian tubes
(C) Bile duct and bronchioles
(D) Fallopian tubes and pancreatic duct

Target Publications

105. In a species, the weight of newborn ranges from 2 to 5 kg. 97% of the newborn with an average weight between 3 to 3.3 kg survive whereas 99% of the infants born with weights from 2 to 2.5 kg or 4.5 to 5 kg die. Which type of selection process is taking place?
(A) Disruptive selection
(B) Cyclical selection
(C) Directional selection
(D) Stabilizing selection

106. The concept of "*Omnis cellula-e-cellula*" regarding cell division was first proposed by:
(A) Schleiden
(B) Aristotle
(C) Rudolf Virchow
(D) Theodore Schwann

107. Identify the CORRECT pair representing the causative agent of typhoid fever and the confirmatory test for typhoid.
(A) *Salmonella typhi* / Anthrone test
(B) *Salmonella typhi* / Widal test
(C) *Plasmodium vivax* / UTI test
(D) *Streptococcus pneumoniae* / Widal test

108. Concanavalin A is:
(A) a lectin (B) a pigment
(C) an alkaloid (D) an essential oil

109. Select the INCORRECT statement.
(A) Inbreeding selects harmful recessive genes that reduce fertility and productivity.
(B) Inbreeding helps in accumulation of superior genes and elimination of undesirable genes.
(C) Inbreeding increases homozygosity
(D) Inbreeding is essential to evolve pure lines in any animal.

110. The CORRECT sequence of phases of cell cycle is:
(A) $S \rightarrow G_1 \rightarrow G_2 \rightarrow M$
(B) $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$
(C) $M \rightarrow G_1 \rightarrow G_2 \rightarrow S$
(D) $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$

111. Which of the following statements is INCORRECT?
(A) Infective constituent in viruses is the protein coat.
(B) Prions consist of abnormally folded proteins.
(C) Viroids lack a protein coat.
(D) Viruses are obligate parasites.

112. *Pinus* seed cannot germinate and establish without fungal association. This is because:
(A) it has very hard seed coat.
(B) its seeds contain inhibitors that prevent germination.
(C) its embryo is immature.
(D) it has obligate association with mycorrhizae.

113. Following statements describe the characteristics of the enzyme restriction endonuclease. Identify the INCORRECT statement.
(A) The enzyme cuts the sugar – phosphate backbone at specific sites on each strand.
(B) The enzyme recognizes a specific palindromic nucleotide sequence in the DNA.
(C) The enzyme cuts DNA molecule at identified position within the DNA.
(D) The enzyme binds DNA at specific sites and cuts only one of the two strands.

114. *Thiobacillus* is a group of bacteria helpful in carrying out:
(A) Nitrification
(B) Denitrification
(C) Nitrogen fixation
(D) Chemoautotrophic fixation

115. What triggers activation of protoxin to active Bt toxin of the *Bacillus thuringiensis* in boll worm?
(A) Alkaline pH of gut
(B) Acidic pH of stomach
(C) Body temperature
(D) Moist surface of midgut

116. Which of the following statements regarding mitochondria is INCORRECT?
(A) Inner membrane is convoluted with infoldings.
(B) Mitochondrial matrix contains single circular DNA molecule and ribosomes.
(C) Outer membrane is permeable to monomers of carbohydrates, fats and proteins.
(D) Enzymes of electron transport are embedded in outer membrane.

117. Which of the following ecological pyramids is generally inverted?
(A) Pyramid of biomass in a forest
(B) Pyramid of biomass in a sea
(C) Pyramid of numbers in grassland
(D) Pyramid of energy

118. It takes very long time for pineapple plants to produce flowers. Which combination of hormones can be applied to artificially induce flowering in pineapple plants throughout the year to increase yield?
(A) Gibberellin and Abscisic acid
(B) Cytokinin and Abscisic acid
(C) Auxin and Ethylene
(D) Gibberellin and Cytokinin

Target Publications

119. Which of the following is the most important cause for animals and plants being driven to extinction?
(A) Economic exploitation
(B) Alien species invasion
(C) Habitat loss and fragmentation
(D) Drought and floods

120. Match the following structures with their respective location in organs:

i.	Crypts of Lieberkuhn	p.	Pancreas
ii.	Glisson's Capsule	q.	Duodenum
iii.	Islets of Langerhans	r.	Small intestine
iv.	Brunner's glands	s.	Liver

Select the correct option from the following:
(A) i – r, ii – s, iii – p, iv – q
(B) i – r, ii – q, iii – p, iv – s
(C) i – r, ii – p, iii – q, iv – s
(D) i – q, ii – s, iii – p, iv – r

121. Polyblend, a fine powder of recycled modified plastic, has proved to be a good material for:
(A) construction of roads
(B) making tubes and pipes
(C) making plastic sacks
(D) use as a fertilizer

122. Tidal Volume and Expiratory Reserve Volume of an athlete is 500 mL and 1000 mL respectively. What will be his Expiratory Capacity if the Residual volume is 1200 mL?
(A) 2200 mL (B) 2700 mL
(C) 1500 mL (D) 1700 mL

123. Persistent nucellus in the seed is known as:
(A) Hilum (B) Tegmen
(C) Chalaza (D) Perisperm

124. What would be the heart rate of a person if the cardiac output is 5L., blood volume in the ventricles at the end of diastole is 100 mL and at the end of ventricular systole is 50mL?
(A) 100 beats per minute
(B) 125 beats per minute
(C) 50 beats per minute
(D) 75 beats per minute

125. The frequency of recombination between gene pairs on the same chromosome as a measure of distance between genes was explained by:
(A) Alfred Sturtevant
(B) Sutton Boveri
(C) T.H. Morgan
(D) Gregor J. Mendel

126. Which of the following factors is responsible for the formation of concentrated urine?
(A) Secretion of erythropoietin by Juxtaglomerular complex.
(B) Hydrostatic pressure during glomerular filtration.
(C) Low levels of antidiuretic hormone.
(D) Maintaining hyperosmolarity towards inner medullary interstitium in the kidneys.

127. Which of the following features of genetic code does allow bacteria to produce human insulin by recombinant DNA technology?
(A) Genetic code is nearly universal
(B) Genetic code is specific
(C) Genetic code is not ambiguous
(D) Genetic Code is redundant

128. Select the correct option.
(A) 11th and 12th pairs of ribs are connected to the sternum with the help of hyaline cartilage.
(B) Each rib is a flat thin bone and all the ribs are connected dorsally to the thoracic vertebrae and ventrally to the sternum.
(C) There are seven pairs of vertebrosternal, three pairs of vertebrochondral and two pairs of vertebral ribs.
(D) 8th, 9th and 10th pairs of ribs articulate directly with the sternum.

129. Variations caused by mutation, as proposed by Hugo de Vries, are:
(A) small and directional
(B) small and directionless
(C) random and directional
(D) random and directionless

130. Which part of the brain is responsible for thermoregulation?
(A) Corpus callosum
(B) Medulla oblongata
(C) Cerebrum
(D) Hypothalamus

131. Colostrum, the yellowish fluid, secreted by mother during the initial days of lactation is very essential to impart immunity to the newborn infants because it contains:
(A) Macrophages
(B) Immunoglobulin A
(C) Natural killer cells
(D) Monocytes

132. How does steroid hormone influence the cellular activities?
(A) Activating cyclic AMP located on the cell membrane.
(B) Using aquaporin channels as second messenger.
(C) Changing the permeability of the cell membrane.
(D) Binding to DNA and forming a gene - hormone complex.

133. Which of the following is a commercial blood cholesterol lowering agent?
(A) Streptokinase (B) Lipases
(C) Cyclosporin A (D) Statin

Target Publications

134. Match Column – I with Column – II.

	Column – I		Column – II
i.	Saprophyte	p.	Symbiotic association of fungi with plants roots
ii.	Parasite	q.	Decomposition of dead organic materials
iii.	Lichens	r.	Living on living plants or animals
iv.	Mycorrhiza	s.	Symbiotic association of algae and fungi

Choose the correct answer from the options given below:
(A) i – q, ii – p, iii – r, iv – s
(B) i – q, ii – r, iii – s, iv – p
(C) i – p, ii – q, iii – r, iv – s
(D) i – r, ii – q, iii – p, iv – s

135. Which of the following protocols did aim for reducing emission of chlorofluorocarbons into the atmosphere?
(A) Gothenburg Protocol
(B) Geneva Protocol
(C) Montreal Protocol
(D) Kyoto Protocol

136. Grass leaves curl inwards during very dry weather. Select the most appropriate reason from the following:
(A) Shrinkage of air spaces in spongy mesophyll
(B) Tyloses in vessels
(C) Closure of stomata
(D) Flaccidity of bulliform cells

137. Which one of the following is NOT a method of *in situ* conservation of biodiversity?
(A) Botanical garden
(B) Sacred grove
(C) Biosphere reserve
(D) Wildlife sanctuary

138. Select the CORRECT sequence of organs in the alimentary canal of cockroach starting from mouth:
(A) Pharynx → Oesophagus → Gizzard → Ileum → Crop → Colon → Rectum
(B) Pharynx → Oesophagus → Ileum → Crop → Gizzard → Colon → Rectum
(C) Pharynx → Oesophagus → Crop → Gizzard → Ileum → Colon → Rectum
(D) Pharynx → Oesophagus → Gizzard → Crop → Ileum → Colon → Rectum

139. Which of the following is TRUE for Golden rice?
(A) It is drought tolerant, developed using *Agrobacterium* vector.
(B) It has yellow grains, because of a gene introduced from a primitive variety of rice.
(C) It is Vitamin A enriched, with a gene from daffodil.
(D) It is pest resistant, with a gene from *Bacillus thuringiensis*.

140. Which of the following glucose transporters is insulin - dependent?
(A) GLUT III (B) GLUT IV
(C) GLUT I (D) GLUT II

141. DNA precipitation out of a mixture of biomolecules can be achieved by treatment with:
(A) Methanol at room temperature
(B) Chilled chloroform
(C) Isopropanol
(D) Chilled ethanol

142. Cells in G_0 phase;
(A) suspend the cell cycle
(B) terminate the cell cycle
(C) exit the cell cycle
(D) enter the cell cycle

143. Select the CORRECT group of biocontrol agents.
(A) *Oscillatoria, Rhizobium, Trichoderma*
(B) *Nostoc, Azospirillium, Nucleopolyhedrovirus*
(C) *Bacillus thuringiensis, Tobacco mosaic virus,* Aphids
(D) *Trichoderma, Baculovirus, Bacillus thuringiensis*

144. Xylem translocates:
(A) Water, mineral salts and some organic nitrogen only
(B) Water, mineral salts, some organic nitrogen and hormones
(C) Water only
(D) Water and mineral salts only

145. Drug called 'Heroin' is synthesized by:
(A) glycosylation of morphine
(B) nitration of morphine
(C) methylation of morphine
(D) acetylation of morphine

146. Respiratory Quotient (RQ) value of tripalmitin is
(A) 0.7 (B) 0.07 (C) 0.09 (D) 0.9

147. Match the hominids with their correct brain size:

	Column I		Column II
i.	Homo habilis	p.	900 cc
ii.	Homo neanderthalensis	q.	1350 cc
iii.	*Homo erectus*	r.	650 – 800 cc
iv	*Homo sapiens*	s.	1400 cc

Select the correct option.
(A) i – r, ii – s, iii – p, iv – q
(B) i – s, ii – r, iii – p, iv – q
(C) i – r, ii – p, iii – s, iv – q
(D) i – r, ii – q, iii – p, iv – s

Target Publications

148. What is the site of perception of photoperiod necessary for induction of flowering in plants?
(A) Shoot apex (B) Leaves
(C) Lateral buds (D) Pulvinus

149. Under which of the following conditions will there be no change in the reading frame of following mRNA?
5′AACAGCGGUGCUAUU 3′
(A) Insertion of A and G at 4th and 5th positions respectively
(B) Deletion of GGU from 7th, 8th and 9th positions
(C) Insertion of G at 5th position
(D) Deletion of G from 5th position

150. Which of the following pair of organelles does NOT contain DNA?
(A) Lysosomes and vacuoles
(B) Nuclear envelope and mitochondria
(C) Mitochondria and lysosomes
(D) Chloroplast and vacuoles

151. What is the genetic disorder in which an individual has an overall masculine development, gynaecomastia, and is sterile?
(A) Edward syndrome
(B) Down's syndrome
(C) Turner's syndrome
(D) Klinefelter's syndrome

152. Identify the cells whose secretion protects the lining of gastro-intestinal tract from various enzymes.
(A) Oxyntic cells (B) Duodenal cells
(C) Chief cells (D) Goblet cells

153. Which of the following contraceptive methods do involve a role of hormone?
(A) CuT, pills, emergency contraceptives
(B) Pills, emergency contraceptives, barrier methods
(C) Lactational amenorrhea, pills, emergency contraceptives
(D) Barrier method, lactational amenorrhea, pills

154. Which of the following statements is NOT CORRECT?
(A) Lysosomes are membrane bound structures.
(B) Lysosomes are formed by the process of packaging in the endoplasmic reticulum.
(C) Lysosomes have numerous hydrolytic enzymes.
(D) The hydrolytic enzymes of lysosomes are active under acidic pH.

155. In *Antirrhinum* (Snapdragon), a red flower was crossed with a white flower and in F_1 generation pink flowers were obtained. When pink flowers were selfed, the F_2 generation showed white, red and pink flowers. Choose the INCORRECT statement from the following:
(A) Ratio of F_2 is $\frac{1}{2}$ (Red): $\frac{2}{4}$ (Pink): $\frac{1}{4}$ (White).
(B) Law of Segregation does not apply in this experiment.
(C) This experiment does not follow the Principle of Dominance.
(D) Pink colour in F_1 is due to incomplete dominance.

156. Match the following hormones with the respective disease:

i.	Insulin	p.	Addison's disease
ii.	Thyroxine	q.	Diabetes insipidus
iii.	Corticoids	r.	Acromegaly
iv.	Growth Hormone	s.	Goitre
		t.	Diabetes mellitus

Select the correct option.
(A) i – t, ii – s, iii – p, iv – r
(B) i – q, ii – s, iii – p, iv – r
(C) i – t, ii – p, iii – q, iv – r
(D) i – q, ii – s, iii – r, iv – p

157. Select the INCORRECT statement.
(A) In domesticated fowls, sex of progeny depends on the type of sperm rather than egg.
(B) Human males have one of their sex - chromosome much shorter than the other.
(C) Male fruit fly is heterogametic.
(D) In male grasshoppers, 50% of sperms have no sex - chromosome.

158. Match the Column – I with Column – II:

	Column I		Column II
i.	P – wave	p.	Depolarisation of ventricles
ii.	QRS complex	q.	Repolarisation of ventricles
iii.	T – wave	r.	Coronary ischemia
iv.	Reduction in the size of T – wave	s.	Depolarisation of atrias
		t.	Repolarisation of atria

Select the correct option.
(A) i – q, ii – p, iii – t, iv – r
(B) i – q, ii – r, iii – t, iv – s
(C) i – s, ii – p, iii – q, iv – r
(D) i – s, ii – p, iii – q, iv – t

159. Match the following genes of the lac operon with their respective products:

	Column I		Column II
i.	i gene	p.	β – galactosidase
ii.	z gene	q.	Permease
iii.	a gene	r.	Repressor
iv.	y gene	s.	Transacetylase

Target Publications

Select the correct option.
(A) i – r, ii – p, iii – s, iv – q
(B) i – r, ii – s, iii – p, iv – q
(C) i – p, ii – r, iii – q, iv – s
(D) i – r, ii – p, iii – q, iv – s

160. What is the direction of movement of sugars in phloem?
(A) Downward
(B) Bi-directional
(C) Non-multidirectional
(D) Upward

161. Which one of the following statements regarding post - fertilization development in flowering plants is INCORRECT?
(A) Central cell develops into endosperm
(B) Ovules develop into embryo sac
(C) Ovary develops into fruit
(D) Zygote develops into embryo

162. Which of the following pairs of gases is mainly responsible for greenhouse effect?
(A) Nitrogen and sulphur dioxide
(B) Carbon dioxide and methane
(C) Ozone and ammonia
(D) Oxygen and nitrogen

163. Which of the following statements is correct?
(A) Cornea consists of dense connective tissue of elastin and can repair itself.
(B) Cornea is convex, transparent layer which is highly vascularised.
(C) Cornea consists of dense matrix of collagen and is the most sensitive portion the eye.
(D) Cornea is an external, transparent and protective proteinacious covering of the eye-ball.

164. Expressed Sequence Tags (ESTs) refers to:
(A) DNA polymorphism
(B) Novel DNA sequences
(C) Genes expressed as RNA
(D) Polypeptide expression

165. Use of an artificial kidney during hemodialysis may result in:
i. Nitrogenous waste build-up in the body
ii. Non-elimination of excess potassium ions
iii. Reduced absorption of calcium ions from gastro intestinal tract
iv. Reduced RBC production
Which of the following options is the most appropriate?
(A) (iii) and (iv) are correct
(B) (i) and (iv) are correct
(C) (i) and (ii) are correct
(D) (ii) and (iii) are correct

166. A gene locus has two alleles A, a. If the frequency of dominant allele A is 0.4, then what will be the frequency of homozygous dominant, heterozygous and homozygous recessive individuals in the population?
(A) 0.16 (AA); 0.48 (Aa); 0.36 (aa)
(B) 0.16 (AA); 0.36 (Aa); 0.48 (aa)
(C) 0.36 (AA); 0.48 (Aa); 0.16 (aa)
(D) 0.16 (AA); 0.24 (Aa); 0.36 (aa)

167. Due to increasing air-borne allergens and pollutants, many people in urban areas are suffering from respiratory disorder causing wheezing due to:
(A) proliferation of fibrous tissues and damage of the alveolar walls.
(B) reduction in the secretion of surfactants by pneumocytes.
(C) benign growth on mucous lining of nasal cavity.
(D) inflammation of bronchi and bronchioles.

168. Which of the following immune responses is responsible for rejection of kidney graft?
(A) Inflammatory immune response
(B) Cell-mediated immune response
(C) Auto-immune response
(D) Humoral immune response

169. Match the following organisms with the products they produce:

	Column I		Column II
i.	*Lactobacillus*	p.	Cheese
ii.	*Saccharomyces cerevisiae*	q.	Curd
iii.	*Aspergillus niger*	r.	Citric Acid
iv.	*Acetobacter aceti*	s.	Bread
		t.	Acetic Acid

Select the correct option.
(A) i – r, ii – s, iii – t, iv – p
(B) i – q, ii – p, iii – r, iv – t
(C) i – q, ii – s, iii – t, iv – r
(D) i – q, ii – s, iii – r, iv – t

170. Consider following features:
i. Organ system level of organisation
ii. Bilateral symmetry
iii. True coelomates with segmentation of body
Select the correct option of animal groups which possess all the above characteristics.
(A) Arthropoda, Mollusca and Chordata
(B) Annelida, Mollusca and Chordata
(C) Annelida, Arthropoda and Chordata
(D) Annelida, Arthropoda and Mollusca

Target Publications

171. Extrusion of second polar body from egg nucleus occurs:
(A) before entry of sperm into ovum
(B) simultaneously with first cleavage
(C) after entry of sperm but before fertilization
(D) after fertilization

172. Which of the statements given below is NOT true about formation of 'annual rings' in trees?
(A) Activity of cambium depends upon variation in climate.
(B) Annual rings are not prominent in trees of temperate region.
(C) Annual ring is a combination of spring wood and autumn wood produced in a year.
(D) Differential activity of cambium causes light and dark bands of tissue - early and late wood respectively.

173. Which of the following sexually transmitted diseases is NOT completely curable?
(A) Genital herpes (B) Chlamydiasis
(C) Gonorrhoea (D) Genital warts

174. The shorter and longer arms of a submetacentric chromosome are referred to as:
(A) q - arm and p - arm respectively
(B) m - arm and n - arm respectively
(C) s - arm and l - arm respectively
(D) p - arm and q - arm respectively

175. Which of the following muscular disorders is inherited?
(A) Myasthenia gravis
(B) Botulism
(C) Tetany
(D) Muscular dystrophy

176. Consider the following statements:
i. Coenzyme or metal ion that is tightly bound to enzyme protein is called prosthetic group.
ii. A complete catalytic active enzyme with its bound prosthetic group is called apoenzyme.
Select the correct option.
(A) Both (i) and (ii) are false.
(B) (i) is false but (ii) is true.
(C) Both (i) and (ii) are true.
(D) (i) is true but (ii) is false.

177. Which of these following methods is the most suitable for disposal of nuclear waste?
(A) Dump the waste within rocks under deep ocean
(B) Bury the waste within rocks deep below the Earth's surface
(C) Shoot the waste into space
(D) Bury the waste under Antarctic ice cover

178. Conversion of glucose to glucose-6- phosphate, the first irreversible reaction of glycolysis, is catalyzed by
(A) Hexokinase
(B) Enolase
(C) Phosphofructokinase
(D) Aldolase

179. Which of the following can be used as a biocontrol agent in the treatment of plant disease?
(A) *Anabaena* (B) *Lactobacillus*
(C) *Trichoderma* (D) *Chlorella*

180. The Earth Summit held in Rio de Janeiro in 1992 was called:
(A) to assess threat posed to native species by invasive weed species.
(B) for immediate steps to discontinue use of CFCs that were damaging the ozone layer.
(C) to reduce CO_2 emissions and global warming.
(D) for conservation of biodiversity and sustainable utilization of its benefits.

Target Publications